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compact1, compact2, compact3

java.lang

Enum Thread.State

java.lang.Object
    java.lang.Enum<Thread.State>
        java.lang.Thread.State

All Implemented Interfaces:

Serializable, Comparable<Thread.State>

Enclosing class:

Thread

```
public static enum Thread.State
extends Enum<Thread.State>
```

A thread state. A thread can be in one of the following states:

- NEW
  A thread that has not yet started is in this state.
- RUNNABLE
  A thread executing in the Java virtual machine is in this state.
- BLOCKED
  A thread that is blocked waiting for a monitor lock is in this state.
- WAITING
  A thread that is waiting indefinitely for another thread to perform a particular action is in this state.
- TIMED_WAITING
  A thread that is waiting for another thread to perform an action for up to a specified waiting time is in this state.
- TERMINATED
  A thread that has exited is in this state.

A thread can be in only one state at a given point in time. These states are virtual machine states which do not reflect any operating system thread states.

Since:

1.5

See Also:

`Thread.getState()`

Enum Constant Summary

Enum Constants

Enum Constant	Description
BLOCKED	Thread state for a thread blocked waiting for a monitor lock.
NEW	Thread state for a thread which has not yet started.
RUNNABLE	Thread state for a runnable thread.
TERMINATED	Thread state for a terminated thread.
TIMED_WAITING	Thread state for a waiting thread with a specified waiting time.
WAITING	Thread state for a waiting thread.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
static **Thread.State**	**valueOf**(**String** name)	Returns the enum constant of this type with the specified name.
static **Thread.State**[]	**values**()	Returns an array containing the constants of this enum type, in the order they are declared.

Methods inherited from class java.lang.Enum

`clone`, `compareTo`, `equals`, `finalize`, `getDeclaringClass`, `hashCode`, `name`, `ordinal`, `toString`, `valueOf`

Methods inherited from class java.lang.Object

`getClass`, `notify`, `notifyAll`, `wait`, `wait`, `wait`

Enum Constant Detail

NEW

```
public static final Thread.State NEW
```

Thread state for a thread which has not yet started.

RUNNABLE

```
public static final Thread.State RUNNABLE
```

Thread state for a runnable thread. A thread in the runnable state is executing in the Java virtual machine but it may be waiting for other resources from the operating system such as processor.

BLOCKED

```
public static final Thread.State BLOCKED
```

Thread state for a thread blocked waiting for a monitor lock. A thread in the blocked state is waiting for a monitor lock to enter a synchronized block/method or reenter a synchronized block/method after calling `Object.wait`.

WAITING

```
public static final Thread.State WAITING
```

Thread state for a waiting thread. A thread is in the waiting state due to calling one of the following methods:

- `Object.wait` with no timeout
- `Thread.join` with no timeout
- `LockSupport.park`

A thread in the waiting state is waiting for another thread to perform a particular action. For example, a thread that has called `Object.wait()` on an object is waiting for another thread to call `Object.notify()` or `Object.notifyAll()` on that object. A thread that has called `Thread.join()` is waiting for a specified thread to terminate.

TIMED_WAITING

```
public static final Thread.State TIMED_WAITING
```

Thread state for a waiting thread with a specified waiting time. A thread is in the timed waiting state due to calling one of the following methods with a specified positive waiting time:

- `Thread.sleep`
- `Object.wait` with timeout
- `Thread.join` with timeout
- `LockSupport.parkNanos`
- `LockSupport.parkUntil`

TERMINATED

```
public static final Thread.State TERMINATED
```

Thread state for a terminated thread. The thread has completed execution.

Method Detail

values

```
public static Thread.State[] values()
```

Returns an array containing the constants of this enum type, in the order they are declared. This method may be used to iterate over the constants as follows:

```
for (Thread.State c : Thread.State.values())
    System.out.println(c);
```

Returns:

an array containing the constants of this enum type, in the order they are declared

valueOf

```
public static Thread.State valueOf(String name)
```

Returns the enum constant of this type with the specified name. The string must match *exactly* an identifier used to declare an enum constant in this type. (Extraneous whitespace characters are not permitted.)

Parameters:

name - the name of the enum constant to be returned.

Returns:

the enum constant with the specified name

Throws:

IllegalArgumentException - if this enum type has no constant with the specified name

NullPointerException - if the argument is null

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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